

Chapter 33. Chronic Ventilator Facilities

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Chronic Ventilator Facilities: Introduction

Chronic ventilator facilities (CVFs) are meant to be “protected” environments for the treatment of patients who require prolonged mechanical ventilation. Numerous words are included under the umbrella of CVFs: long-term acute care facilities, respiratory special care units, chronic ventilator-dependent units, regional weaning centers, ventilator-dependent rehabilitation hospitals, prolonged respiratory care units, noninvasive respiratory care units, high-dependency units, and respiratory intensive care units. And this is without even considering nursing homes and hospice care.

A 16-year study showed that the number of acute care hospital beds in the United States has decreased over time, but the number of critical care beds has increased progressively in both absolute and proportional terms. Indeed, the total number of non-critical care beds decreased by 31%, whereas critical care beds increased by 26%¹; nevertheless, admissions to an intensive care unit (ICU) are very strongly influenced by bed shortages. Most beds in an ICU are occupied by patients requiring mechanical ventilation. A subset of patients receiving mechanical ventilation may have weaning difficulties, so that the duration of ventilation may be abnormally prolonged (commonly defined as greater than 15 days). Several reports indicate that these ICU patients are affected by complex cardiopulmonary disease or multisystem problems and have a relatively poor outcome.^{2–6}

Of 6,469,674 hospitalizations in six American states, 180,326 (2.8%) received invasive mechanical ventilation. A total of 44.6% had at least one major comorbidity condition. The most common comorbidities were diabetes (13.2%) and pulmonary disease (13.2%), and only 30.8% of patients were discharged to home from the hospital, while the others ultimately required care in a skilled care facility.⁷

Increased life expectancy has dramatically increased the age of patients requiring critical care. Medical ICU admission is associated with a high long-term mortality even in healthy elderly patients, while most of the oldest survivors undergo prolonged mechanical ventilation, which is often a marker not only of respiratory system insufficiency, but also a multisystems insufficiency caused by many factors, including a chronic underlying disease, infections, malnutrition, complications, invasive procedures, and medications.^{8,9}

Prolonged mechanical ventilation is not just a “medical” problem; it also has social and economic impact. Costs for mechanical ventilation in the United States are estimated to be \$27 billion, representing 12% of all hospital costs. Incidence, mortality, and cumulative population costs rise significantly with age.⁷ Each year in the United States, approximately 300,000 patients receive prolonged life support in an ICU, and this number is likely to double within a decade, with associated costs of more than \$50 billion.¹⁰

These patients, once discharged from the ICU, have a readmission rate of 67% and spend an average of 74% of all days alive in a hospital or in an acute care facility, or receiving home health care. Indeed, patients who survive for 1 year are left with a serious burden of pervasive and persistent disability despite aggressive care that cost 306,135 \$ for cohort members.¹¹ The burden of prolonged mechanical ventilation also affects the families because of financial implications, disruption of family routine, and uncertainty of the future, as well as community financial resources. Even after these “chronically ill” patients are discharged from the ICU, which occurs once the precipitating cause of their acute episode of respiratory failure has been reversed, they still require mechanical ventilation. Thus, they either require transfer to a long-term care or rehabilitation facility.¹²

Rationale

At the time of ICU discharge, patients who still need dedicated care or mechanical support are usually in a phase of clinical stability, but they have several “open” problems that prevent them being transferred to a regular ward or directly to home. Several reasons justify their

transfer to a dedicated CVF.

First, weaning from prolonged mechanical ventilation is a complex, time-consuming process that involves not only the selection of the best ventilation method for a particular patient, but comprehensive procedures, such as protocol-driven weaning, have the potential for improving clinical outcome through increased use of efficacious weaning methods, such as daily weaning screening, spontaneous breathing trials, and management of the ventilator and sedation.¹³ Rapid turnover in a busy ICU and its clinical burden does not always allow devoting the necessary time for these procedures, and discontinuation of mechanical ventilation is not always a priority.¹⁴

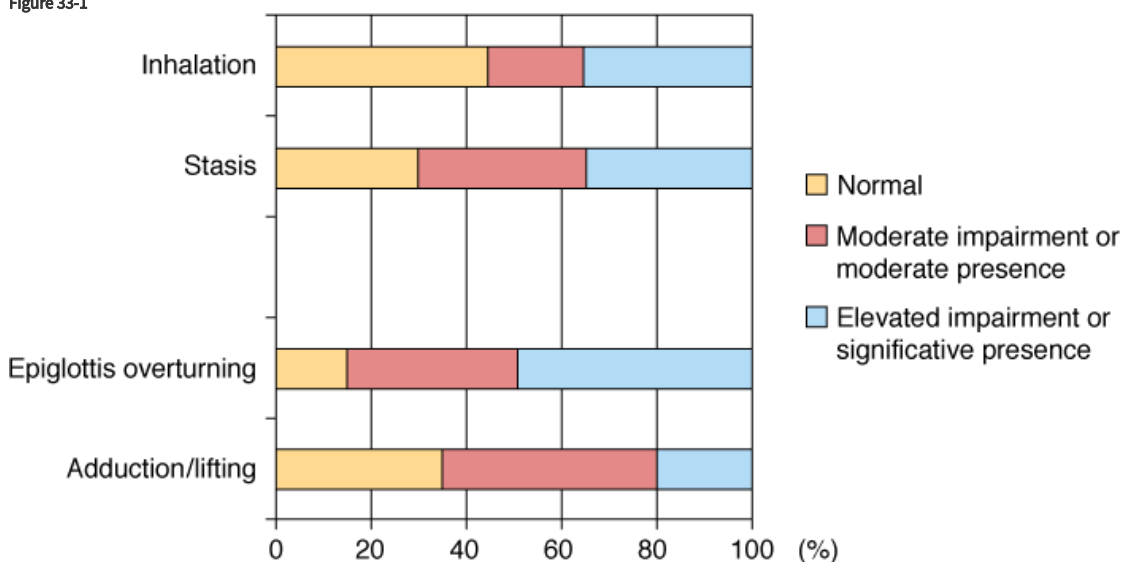
Second, given the nature of critical illnesses and the modalities used to manage them, prolonged bed rest, with well-known adverse physiologic effects, is the rule in the ICU. Rehabilitation has the potential to restore lost function. Yet, with few exceptions,¹⁵ it traditionally does not start until after ICU discharge. Critically ill patients are often viewed as “too sick” to tolerate physical activity in the early phase of their illness, frequently prolonging immobilization.

Third, psychiatric symptoms are quite common in chronic patients who survive an ICU stay and in their relatives at the time of ICU discharge, as well as 90 days later.¹⁶ Open visiting hours for family members and comfort among patients and caregivers may be enhanced by return to a more physiologic circadian rhythm as opposed to that found in a typical ICU. Delirium, probably the most common psychiatric problem in critically ill patients, may still occur after discharge from an ICU, albeit with a lower incidence. The environment and the strategy of care adopted in CVFs seems to positively affect the recovery from a delirious state.¹⁷

Fourth, sleep disruption is a well-documented problem during ICU stay and it persists at ICU discharge.¹⁸ Sleep deprivation can have significant consequences and has been shown to impair cognitive function, increase protein catabolism, decrease immune function, and alter respiratory mechanics that could eventually impact weaning from mechanical ventilation. A more “natural setting,” like a CVF, may minimize the number and intensity of some environmental factors, involved in the genesis of poor sleep, such as noise and light intensity.¹⁹

Fifth, aspiration is common in patients with tracheostomies receiving prolonged ventilation and with advanced age the risk of aspiration increases. Swallowing problems are seldom assessed with the appropriate technique (i.e., videofluoroscopy) in the ICU, and the most important specific rehabilitative therapy is rarely started. Figure 33-1 illustrates the incidence and types of swallowing dysfunction in a cohort of 188 patients admitted to a CVF after ICU discharge.

Figure 33-1



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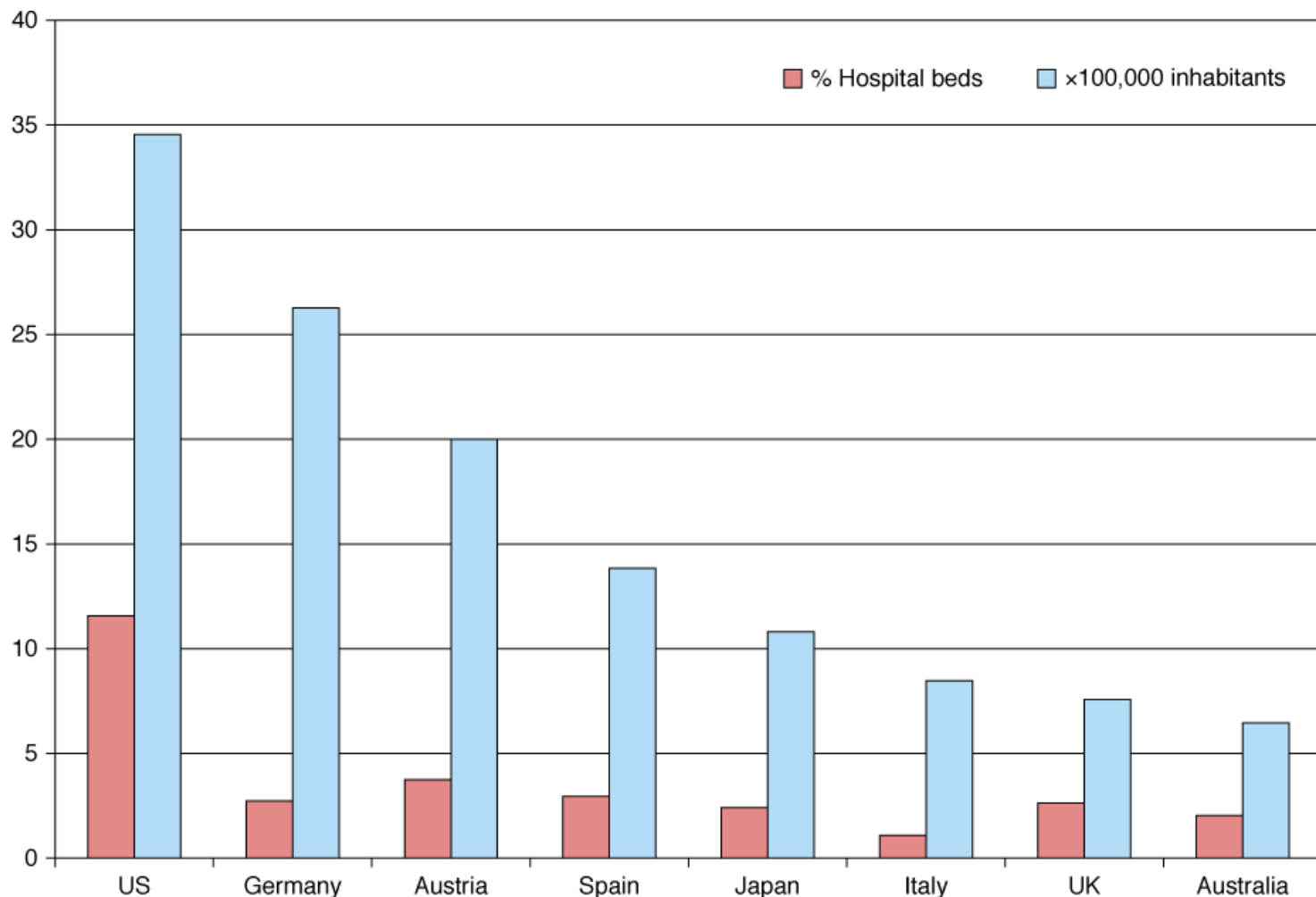
Incidence and types of swallowing dysfunction in a cohort of 188 patients admitted to a chronic ventilator facility after discharge from an ICU (Ceriana et al, unpublished data). Epiglottic overturning refers to the backward movement of the epiglottis during closure of the laryngeal vestibule. Adduction/lifting refers to the adduction of the vocal folds and movement of the larynx observed during inspection when the patient is requested to phonate and swallow.

Sixth, the number of tracheotomies has dramatically increased over the last few years, not only because of increasing numbers of difficult-to-wean patients, but also because of early timing of the procedure. The vast majority of these patients are discharged with a tracheal tube still in place, even when they no longer need ventilator support. A Spanish study shows that lack of tracheostomy decannulation in the ICU is associated with markedly increased ward mortality, at least in a large subset of these patients. The increased ward mortality may have resulted because the care of a tracheotomy is suboptimal outside a specialized environment.²⁰ Ceriana et al²¹ highlighted the possibility of safely removing the tracheotomy cannula from almost 80% of patients breathing autonomously among a group of seventy-two patients recovering from weaning in a CVF, while decannulation was achieved in approximately 40% of patients transferred to a similar environment in the United States.²²

Seventh, the interaction between ICU physicians and patients or their surrogates is far from optimal, especially when discussing the problem of prolonged ventilation including its outcome. A lack of prognostication, discordance between surrogates and physicians about potential outcomes, and unreasonably optimistic expectations of surrogates are potentially modifiable factors in surrogate–physician interactions. A less stressful environment than the ICU, combined with fewer sedative drugs and more comprehensive teamwork—nurses, respiratory therapists, psychologists, and clergy—typical of a CVF, may improve the patient–clinician relationship so that expectations and outcomes may be better defined and discussed.²³

Overall, the disproportionate number of days of care required by chronically ventilated patients contributes to overcrowding despite the limited availability of ICU beds.²⁴ As Figure 33-2 shows the number of ICU beds, in several countries. The demand for ICU beds far exceeds their availability in many European and non-European countries, so that many patients are denied ICU admission because the ICU beds are already fully occupied.²⁵ Consequently, many critically ill patients are cared for in non-ICU beds throughout the hospital. The outcome of patients who fit ICU admission criteria but who are hospitalized in regular wards was assessed in a study performed in Israel.²⁶ Early survival advantage in an ICU suggests that a window of critical opportunity exists for these patients. Given economic constraints and the dearth of ICU beds, it is possible that increased turnover of patients in an ICU, thus providing an opportunity for more needy patients to avail of the early benefits of intensive management, may be advantageous. This study highlights both the risk of not admitting critically ill patients to the ICU and the advantages of early discharge from the ICU for the subset of patients not in immediate life-threatening situations, but yet not ready for discharge to a nonprotected environment.²⁶

Figure 33-2



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Numbers of actual ICU beds as percentage of hospital beds and as per 100,000 inhabitants.

The paucity of ICU beds is not as dramatic in the United States as in Europe, but discharge of chronic ventilator patients is still problematic. There has been an increase in the number of facilities specializing in the care of chronically ventilated patients. In Massachusetts, for example, the estimated prevalence of prolonged ventilation increased from 2.8 per 100,000 inhabitants in 1983 to 7.1 per 100,000 inhabitants in 2006, and most of these ventilator-dependent patients are in long-term acute care facilities, large urban teaching hospitals, and at home. The number of long-term acute care hospitals in the United States increased at a mean rate of approximately 9% per year, from 192 in 1997 to 408 in 2006, which resulted in an increased incidence of long-term acute care utilization after critical illness from 38.1 per 100,000 in 1997 to 99.7 per 100,000 in 2006.²⁷

Organization of Chronic Ventilator Facilities

The term *CVF* is not necessarily synonymous with the so-called intermediate care unit or high-dependency unit, which are meant for patients who do not require full ICU care but are thought to need more care than usually can be offered in a general ward.^{28–32} Not all patients in the latter units may be ventilator-dependent, but they may require either noninvasive monitoring or noninvasive mechanical ventilation. There is no agreement about the classification of facilities for patients needing prolonged mechanical ventilation because of geographic differences, lack of consensus regarding the appropriate timing of transfer from the ICU, and different criteria of admission. For example, the American College of Critical Care Medicine states in its guidelines for admission to and discharge from adult intermediate care units that “medically stable ventilator patients for weaning and chronic care” are the ideal candidates for these environments.³³ Unfortunately, these units were described only generically as “progressive-care units or single-organ subspecialty floors or chronic ventilator respiratory-care units.” Details were not provided on how these units should be organized or financially reimbursed.

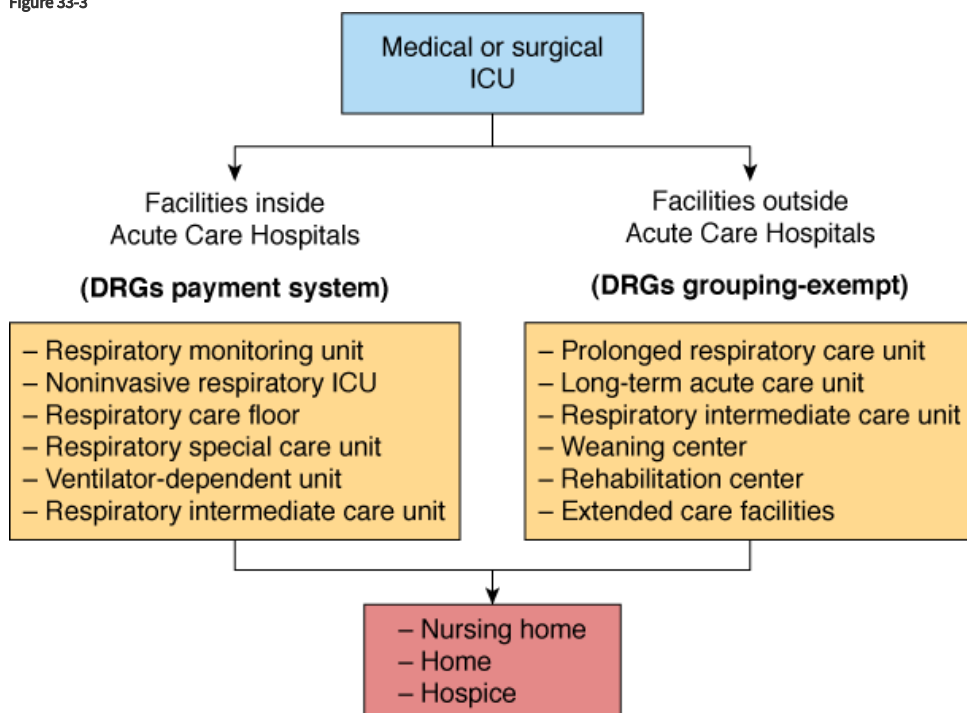
Timing of discharge from the ICU is also critical. Long-term ventilator patients are often old (i.e. >75 years) and have various underlying chronic comorbidities that may complicate or exacerbate their respiratory condition at any time after discharge from the ICU.^{34,35} The 6-month rate of readmission to an acute care hospital is close to 40%, and readmission is often within the first 2 months after discharge from the ICU.³⁶ Surprisingly, this rate is not influenced by the initial discharge disposition; that is, it does not differ for patients discharged to a nursing home, a CVF, or their own home. These findings indirectly suggest that not all CVFs are presently prepared to [cope](#) with the burden of a new “acute exacerbation” in such patients. For example, Nasraway et al³⁷ found that despite a 31-fold increase in the number of all adults transferred from ICUs to extended care facilities in the Boston area between 1990 and 1996, the level of care of these facilities varied greatly depending mainly on the availability of skilled nurses.

There is still disagreement about the definition of a ventilator-dependent patient. The ninth revision of the *International Classification of Diseases*³⁸ defines long-term ventilation patients as those who have received 5 or more days of ventilation. Various authors, however, have used limits as short as 48 to 72 hours and as long as 40 days.^{39–41} Realistically, approximately 20% of patients in an ICU require mechanical ventilation for more than 1 week, and about half of them are weaned successfully over the following few days.⁴² Therefore, a limit of 2 weeks has been chosen by most authors to define the threshold for ventilator dependency. The Health Care Financing Administration⁴³ has expanded this limit to 21 days of mechanical ventilation for at least 6 hours a day. A definition based only on time, however, does not consider that for a particular patient to be regarded as ventilator-dependent (and therefore eligible for transfer to a chronic care facility), the precipitating cause of the respiratory failure must have been reversed.

CVFs have been described in the literature only in North America,⁴⁴ Europe,⁴⁵ and Asia.⁴⁶ Substantial differences exist in their organization, location, and criteria of admission. An editorial entitled, “The Challenge of Prolonged Mechanical Ventilation: A Shared Global Experience,”⁴⁷ stressed the need for common international consensus. Yet international guidelines and/or “position papers” are lacking.

Given the confusion of terminology, we submit that the most logical classification of CVFs is one based on the location of the different facilities, specifically whether the facility is inside or outside a so-called acute care hospital. [Figure 33-3](#) illustrates the possible sites of care for patients who are chronically ventilator-dependent. It should be borne in mind that access to these different environments may differ internationally or even regionally within the same country.

Figure 33-3



Potential sites of care for ventilator-dependent patients when discharged from an intensive care unit (ICU). *DRG*, diagnostic-related grouping.

Facilities Within Acute Care Hospitals

Mechanical ventilation is initiated outside the hospital in a relatively small proportion of patients; ventilation is mostly started and stabilized in an ICU.⁴⁸ The timing of discharge of ventilator-dependent patients from the ICU is linked strictly to the criteria of admission of each single CVF, which are described in [Chapter 34](#).

In the late 1980s, CVFs started to emerge in acute care hospitals as an attempt to provide an alternative therapeutic environment for ICU patients requiring prolonged mechanical ventilation. Krieger et al^{49,50} probably were the first to establish a CVF within an acute care hospital (Central Respiratory Monitoring Unit at Mount Sinai Medical Center, Miami, FL). They were followed shortly after by Elpern et al^{51,52} (Noninvasive Respiratory Care Unit of St. Luke's Medical Center, Chicago, IL). These units were devoted mainly to patients requiring prolonged mechanical ventilation. In 1990, the Mayo Clinic opened a ventilator-dependent unit inside Saint Mary's Hospital.^{53–56} Its mission was to create an environment conducive to the rehabilitation of patients with respiratory failure and also to lower the costs of ventilator-dependent patients.⁵⁷ Shortly thereafter, new CVFs were opened. These went under different names at Saint Vincent's Hospital and Medical Center in New York, (NY)⁵⁸ (Nonmonitored Respiratory Care Floor) and at Cleveland Clinic Foundation in Cleveland, Ohio⁵⁹ (Respiratory Special Care Unit). As discussed later (see "[Outcomes and Effectiveness](#)"), the overall rate of weaning success was high (>50%), with a mortality rate below 40%.

In Europe, the first report on this subject was published in 1995 by Smith and Shneerson⁶⁰ from England. Rather than opening a special unit for patients requiring prolonged ventilation, they described the institution of a progressive-care multidisciplinary program carried out within the ICU by a dedicated team of respiratory physicians and nurses. As soon as patients were judged ready for discharge from the ICU, they were transferred to a respiratory unit, where they continued the multidisciplinary program. The program was very successful with regard to discharge to home (80% of the patients) and 1-year survival (76%). A survey⁴⁵ of the European Respiratory Society on respiratory intermediate care units (RICUs) showed that during 1999, most (58%) of the 11,890 patients admitted to fifty-five RICUs received invasive mechanical ventilation. These units almost always (>90%) were inside an acute care hospital. Unfortunately, no data were available on how many of these patients actually were ventilator-dependent. Nevertheless, an Italian survey published in 2001⁶¹ reported that 61% of patients receiving invasive mechanical ventilation in Italian RICUs were tracheotomized and therefore considered ventilator-dependent. The percentage is similar to that reported in Britain and Germany, where most patients survived to leave hospital, most having been weaned from the ventilator. Survivors were younger and spent less time ventilated in the referring ICU.^{62,63}

Irrespective of location on either side of the Atlantic, the organization and staffing of CVFs inside acute care hospitals appear to be homogeneous and more similar to that of an ICU than to that of a CVF located outside an acute care hospital.^{64,65} Most of these CVFs provide noninvasive monitoring, the nurse-to-patient ratio is usually 1:2 to 1:4,^{45,51,53,58,59} and a lead respiratory therapist is assigned permanently and is present in the unit. General medical care is provided around the clock by medical house staff under the direction of an attending physician in either critical care or pulmonology. The nursing staff usually is specially trained through orientation and in-service programs to address the needs of this particular patient population. The approach to the patients is multidisciplinary, involving dietitians, psychologists, physical therapists, speech therapists, social workers, and clergy as needed. Because these CVFs are located within an acute care hospital, all the diagnostic (e.g., computed tomographic scans, nuclear magnetic resonance imaging) and therapeutic (e.g., major surgery) options are readily available. In Europe, the nurse-to-patient ratio varies slightly according to the three levels of care of RICUs.^{29,45} Because of the different education and responsibilities of the respiratory therapists (rarely present around the clock), an attending physician is always present in the units.^{66,67}

Facilities Outside Acute Care Hospitals

This classification consists of several CVFs going under names such as regional weaning centers, prolonged respiratory care units, long-term acute care units, and RICUs inside rehabilitation hospitals. The need for these special facilities outside the acute care health system was recognized a long time ago in North America, but only recently in Europe. For example, the Comprehensive Critical Care of the British

Department of Health⁶⁸ stated that “the effectiveness of specialist weaning and progressive care programs for long-term ventilation of patients has been demonstrated by research, and NHS Trusts should review the need for provision of such services.”

The first experience of a CVF dedicated to the problem of ventilator-dependent patients was reported by Indihar,^{69–71} who described 10 years of activity, starting in 1979, of a unit located in Bethesda Lutheran Medical Center. In the late 1980s there was substantial growth of regional weaning centers in the United States. Examples include the Barlow Respiratory Hospital in Los Angeles,^{72,73} the Medical Center of Central Massachusetts,⁷⁴ and the Hospital for Special Care, New Britain, Connecticut.^{75,76} Later, there was an impressive burgeoning of new long-term acute care units. By 1997, these had a capacity of about 15,000 patients per year.⁷⁷ These units were established either as free-standing hospitals, as in the case of most regional weaning center, or within an acute care hospital but operating with total independence. As such, their governance is independent of the host hospital, and reimbursement is not based on a diagnosis-related grouping (DRG) system.

CVFs within a rehabilitation hospital are also popular in the United States and Europe, especially in Germany and Italy, where approximately 15% of RICUs are located inside rehabilitation centers.^{45,62,78,79}

Unlike CVFs located within acute care hospitals, CVFs located outside appear to have a rather heterogeneous organization and staffing. This heterogeneity occurs despite these facilities espousing a program based on the common ideal of providing comprehensive medical, nursing, and respiratory care to ventilator-dependent patients. For example, New England³⁷ has several “extended care facilities” outside acute hospitals; the skills and level of care vary dramatically among different centers. Despite the personnel being fully licensed health care practitioners, they may not all be completely familiar with the complexities of ventilator-dependent patients. Patient outcome is likely to depend on the different levels of care provided.

In most North American centers, the nurse-to-patient ratio is approximately 1:4 during the day and 1:8 at night and on weekends. A full-time respiratory therapist^{72,80} usually is present. The primary physicians are either internists or specialists in pulmonary and/or critical care medicine, whereas nighttime coverage commonly is provided by junior doctors. The comprehensive care team includes physical therapists, occupational therapists, speech and swallowing therapists, and clinical psychologists. Screening for admission is performed either by an attending physician or by a nurse in consultation with an attending physician.⁸⁰ Weaning protocols and selection of ventilator settings during this process are implemented by respiratory therapists.⁸¹ Discharges are planned by nurses or social-work care managers. With the exception of a few facilities, which may have operating rooms for minor surgery, most CVFs located outside acute care hospitals cannot offer surgery or sophisticated diagnostic procedures.

In Europe, CVFs outside acute care hospitals are run mainly by full-time attending physicians who are specialists in respiratory and/or critical care medicine.^{29,45,61} These physicians are in charge of the admission and discharge of patients and the weaning protocols. They are on duty 24 hours a day. The doctor-to-patient ratio is at least 1:8. The nurse-to-patient ratio is usually similar to that of North American centers. Because of their different educational training, respiratory therapists in Europe tend to be involved mainly in rehabilitation programs, for approximately 8 hours a day (excluding Sundays and holidays), rather than in the weaning process.²¹ In common with North American facilities, most European CVFs do not offer major surgery.

Irrespective of their geographic location, CVFs located outside acute care hospitals are intended to provide privacy, rest, and longer visiting hours for relatives and friends. Above all, they provide physical and pulmonary rehabilitation, which has been shown to help in freeing patients from mechanical ventilation and restoring them to an acceptable level of autonomy.⁸²

Nursing Homes

A small percentage of ventilator-dependent patients are discharged from an ICU³⁶ directly to a nursing home. Nursing homes, however, are more likely to receive such patients once they have left a CVF. In 1991, a survey by the American Association for Respiratory Therapists⁸³ found that nearly 30% of ventilator-dependent patients remained in CVFs for nonmedical reasons, such as reimbursement obstacles to discharge or lack of postdischarge placement options. Indeed, approximately 20% of patients cared for in a ventilator-dependent unit are transferred to a nursing home simply because they are not ready to go home.³⁶

Nursing homes have been established all over the United States,⁸⁴ either as independent units inside larger facilities or as stand-alone facilities. To the best of our knowledge, in Europe specialized units for the care of ventilator-dependent patients are very few.⁸⁵ Apparently, there is no standardization of admission criteria, staffing, or organization of these units apart from the person needing “24-hour nursing care for a cognitive or a physical impairment.”⁸⁶ Nurses working in this specialized area should be trained by respiratory therapists to perform specific procedures such as suctioning, tracheotomy care, and monitoring of ventilator parameters. In some cases, for example, Lakeside Hospital in Wisconsin, Eau Claire⁸⁷ weekly care rounds led by a pulmonary physician with the participation of the care team, including not only the certified nurses but also respiratory therapists, dieticians, social workers, and, when possible, family members, have been introduced. In the very few observational studies performed in this kind of facility, weaning outcomes are very promising.⁸⁷ Further studies are needed to define the characteristics of ventilator-dependent patients who are most likely to benefit from admission to this environment.

Criteria for Admission

Table 33-1 lists criteria for defining the “ideal” candidate for a CVF. The concept of ventilator dependency is the primary criterion for selecting admissions to a CVF. Accordingly, most centers accept only tracheotomized patients because tracheotomy per se is assumed as evidence of ventilator dependency. Significant differences on the best location of care at the time of ICU discharge between patients with and without tracheostomy has been demonstrated.⁸⁸ Indeed, the dramatic increase in tracheotomies performed over the last 10 years⁸⁹ suggests that ICU physicians tend to perform an early tracheotomy before trying to complete weaning. This change may reflect attempts to decongest busy ICUs more rapidly by allowing transfer of ventilator-dependent patients to extended care facilities. To avoid this problem, some units request proof that a patient has failed at least two weaning trials before being admitted.⁵⁵ Other centers, however, accept patients based mainly on the availability of resources (bed and nursing staff) rather than on the basis of perceived ability to wean.⁶²

Table 33-1: Criteria for Defining the Ideal Candidate to a Chronic Ventilator Facility

Patients ventilated for more than 14 days, necessitating prolonged weaning
Presence of a tracheostomy
Potential weaning possibility
Defined diagnosis
Single-organ failure
Hemodynamically stable (no pressor infusion)
No sepsis or active infections
Absence of surgical problems
No need of continuous sedation
Absence of chronic renal failure necessitating hemodialysis
PaO ₂ /FI _O ₂ > 200 and need for external PEEP <10 cm H ₂ O
Need for aggressive rehabilitation program
Potential to benefit from a rehabilitation program

Abbreviations: FI_O₂, fractional inspired oxygen concentration; PaO₂, partial pressure of arterial oxygen; PEEP, positive end-expiratory pressure.

Another criteria used for selecting patients for admission to CVFs are the clinical stability and the potential to benefit from a rehabilitation program.^{53,59} The clinical stability is defined as reversal of the precipitating cause of respiratory failure, hemodynamic stability (not needing invasive measurements of blood pressure or pulmonary artery pressure, not requiring vasoactive drug infusion before transfer) and absence an arrhythmia requiring telemetry. Patients with multiorgan failure often are not admitted. Patients receiving hemodialytic support are usually accepted, particularly if the unit is inside an acute care hospital. When transferring a patient from the ICU, because

errors in discharge reports are frequent, it is necessary to double check to avoid errors. It has been shown, for example, that of 123 transfer reports, seventy-six (62%) were affected by at least one error, with 28% of the errors being potentially harmful to the patients.⁹⁰

The idea of the ability to benefit from rehabilitation or other comprehensive program is subjective and depends strongly on the judgment of the proposing or accepting physician. In some facilities, the proposing physician must make a written statement where the physician states that the patient is either capable of being weaned from the ventilator or is likely to return to the community despite receiving ventilator assistance.⁵⁴ Other facilities specifically ask that a patient be returned to the referring hospital once the patient has been weaned successfully or when it becomes obvious that weaning will not be possible. To ensure that all parties understand this policy, a letter of confirmation may be required from the referring physician, countersigned by the patient or the patient’s surrogate.⁹¹ A more liberal attitude is shown by other facilities, for example, the Worcester County Ventilator Unit, Worcester, Mass⁷⁴ where “specifically no admission decision is based on prognosis, weaning or rehabilitation potential.” Also, at the Long-Term Acute Care Unit of the University of Chicago,⁸⁰ “patients are accepted regardless of the severity of their illness, provided they are stable for transfer.” The latter institution theoretically accepts patients requiring positive end-expiratory pressure of more than 10 cm H₂O or a fractional inspired oxygen concentration (F_IO₂) of greater than 60%. Finally, methods have been proposed for identifying patients suitable for admission to CVFs based on severity-of-illness scores or activity of treatments.^{92–95}

With few local differences, the classification of ventilator-dependent patients admitted to CVFs follows that illustrated⁹⁶ in Table 33-2.

Table 33-2: Classification of Patients Admitted to a Chronic Ventilator Facility

Acute lung injury
Acute respiratory distress syndrome
Pneumonia
Aspiration injury
Burns
Chronic lung disease
Chronic obstructive pulmonary disease
Asthma
Pulmonary fibrosis
Fibrothorax
Postoperative
Lobectomy or pneumectomy
Cardiac surgery
Major abdominal surgery
Neuromuscular diseases
Amyotrophic lateral sclerosis
Multiple sclerosis
Postpolio syndrome
Spinal cord injuries
Kyphoscoliosis
Critically ill polyneuropathy
Guillain-Barré syndrome
Cardiovascular disorders
Chronic congestive heart failure
Ischemic cardiomyopathy

Comparisons with Intensive Care Units

Commenting on the rationale for opening an intermediate care unit, Vincent and Buchardi⁹⁷ stated they “would recommend that instead of fragmenting ICU facilities by separating ‘intensive’ from ‘intermediate,’ with the potential risk of reduced staff morale and less adequate patient care, intensive and intermediate care beds should be combined in one unit.” As stated earlier, intermediate care units are not necessarily equivalent to CVFs. Despite the lack of strong scientific evidence, specific facilities for ventilator-dependent patients do have some clinical and financial advantages over ICU care (they are not, however, “in competition”).^{31,98–101} Table 33-3 lists the main differences between ICUs and CVFs located inside or outside acute care hospitals.

Table 33-3: Differences in Services Provided by Intensive Care Units and Chronic Ventilator Facilities

	ICU	CVFs Inside an Acute Care Hospital	CVFs Outside an Acute Care Hospital
Medical-centered care	+++	++	+
Dedicated multidisciplinary team	+	+++	++++
Nurse-to-patient ratio	+++	++	++
Invasive monitoring	+++	++	+
Diagnostic availability	+++	+++	++
Privacy	+	++	+++
Family contact	+/-	++	++++
Noise and artificial light	+++	++	++
Surgical availability	+++	+++	+
Hemodialysis	+++	+++	+
Physiotherapy and rehabilitation	+	+++	++++
Costs	+++	++	++

Abbreviations: + + + +, High; + + +, Quite High; ++, Moderate; +, Low; +/-, Poor.

A retrospective review¹⁰¹ of 429 ICU patients tracheotomized for respiratory failure and needing prolonged mechanical ventilation showed that only 57% of survivors had been free from mechanical ventilation by the time of discharge from the ICU. Patients who were finally weaned and had their tracheostomy tubes removed had better survival than patients who did not. Improved survival, however, came at higher hospital cost and a longer ICU stay. This study illustrated the need to compare the clinical outcomes and financial burdens of these “difficulties to wean” patients when treated exclusively in a traditional ICU or transferred during hospitalization to a CVF. The study also demonstrated that the emotional status of these patients at ICU discharge was generally good, whereas physical function was quite limited.¹⁰¹

The only randomized, controlled study directly comparing the outcomes of chronically ill patients, most of them ventilator-dependent, managed entirely in the ICU or transferred when clinically stable to a special care unit (SCU), managed mainly by specialized nurses, was performed by Rudy et al.¹⁰² The SCU had a case-management approach for clinical problems, a full rehabilitation program, weaning protocols, and control of resource use. A total of 220 patients were assigned randomly to either the SCU or the ICU. Overall mortality rates

were similar in the two groups, but the rate of readmission was lower and hospital stay shorter in the SCU group. The average cost of delivering care was \$5000 less per patient in the SCU than in the ICU, and the cost to produce a survivor was \$19,000 less. It makes sense that specialized care may achieve better outcomes. It is difficult, however, to draw up standardized recommendations based on a relatively small single-center study where the two groups were not fully comparable in either number or baseline conditions.

Using a retrospective chart review and questionnaires, Seneff et al¹⁰³ analyzed 6-month mortality and hospital costs in fifty-four acute care referral hospitals and twenty-six long-term acute care institutions. Hospital costs included the amount of uncompensated care incurred by the ICU under the Medicare prospective payment DRG system. The authors compared 432 patients ventilated for an average of 3 weeks who were referred but not transferred to CVFs (Vencor) with 1702 patients who were referred and transferred to CVFs. Six-month mortality was not adversely affected by transfer to a CVF. Because patients had long hospital stays and consumed much resources, overall cost of care was very high. Acute care hospitals, however, theoretically can reduce the amount of uncompensated care for these patients by timely referral to an appropriate CVF. Only approximately 10% of the two groups were discharged directly home; most patients were transferred to a nursing home or another CVF.

The latter study¹⁰³ leaves three major unsolved issues. First, the “appropriate” timing of transfer has not been clearly defined because some CVFs accept sicker patients with “overt” nonpulmonary dysfunction, whereas others only accept patients with the “single-organ failure.” Second, although the authors showed that transfer to a CVF is associated with a reduction of ICU uncompensated costs of care, the issue of best reimbursement system for the CVF was not assessed. Third, even after patients were transferred to a CVF, the rate of discharge home was disappointingly low; there is an urgent need to identify the best location of care for these patients once they are discharged from a CVF.

One of the most important rationales for using a CVF is the expected reduction in ICU stay. Otherwise, the marginal variable cost of the ICU is relatively small compared with the average total cost.¹⁰⁴ For this reason, interventions that reduce ICU length of stay and duration of mechanical ventilation may not result in significant actual cost savings.¹⁰⁴

Observational Studies

The patient populations admitted to CVF, as reported by recent observational studies from the United States and Europe, span a wide range of ages and admission diagnoses, and generally have significant comorbidities. Table 33-4 summarizes outcome indices for patients admitted to an acute care setting for weaning. Although comparisons between studies is difficult, the rate of weaning success ranges from 25% to greater than 90%. Variability of hospital mortality is equally great, ranging from 10% to 50%.

Table 33-4: Outcomes for Ventilator-Dependent Patients Admitted to a Chronic Ventilatory Facility Within an Acute Care Hospital

Auhor (Ref)	Year	Patients (n)	Age (years)	Patients	Patients Weaned (%)	Patients Died in Hospital (%)	Patients Died within 1 Year (%)	Duration of MV (days)	Postdischarge Location (%)
Elpern ⁵¹	1989	95	71.6	Mixed	31%	67%	?	8.1	?
Rudy ¹⁰²	1995	145	64 ± 12	Mixed	?	44%	?	?	?
Latriano ⁵⁸	1996	224	66 ± 17	Mixed	51% (all patients) 92% (survived)	50%	?	50 ± 66	H = 31% CVFo = 64% Other = 5%
Douglas ³	1997	57	61 ± 20	Mixed	?	44%	50%	28	H = 33% CVFo = 42% NH = 18% Other = 7%
Gracey ⁵⁵	1997	206	65 ± 14	Mixed	74%	8%	31%	?	?
Dasgupta ⁵⁹	1999	212	68	Mixed	60%	18%	?	17	H = 34% Other = 66%
Robson ⁹⁸	2003	161	69	Mixed	89%	14%	?	8	?
Engoren ¹⁰¹	2004	429	68	Mixed	57%	22%	36%	29	?
Engoren ¹⁰⁵	2005	113	47 ± 21	Trauma patients	75%	2%	81%	25 ± 21	H = 5% CVFo = 5%; Rehabilitation = 32% Other = 6%
Kahn ²⁷	2010	244,621	76.9	Mixed	?	25%	50.4%	?	H = 30.73% Acute care = 17.63% Other = 26.4%

Abbreviations: CVFo, chronic ventilator facility outside an acute care hospital; H, home; MV, mechanical ventilation; NH, nursing home hospital.

A U.S. study showed that, over time, patients transferred to CVFs had a higher number of comorbidities (5 in 1997 to 2000 vs. 5.8 in 2004 to 2006) and were more likely to receive mechanical ventilation at the long-term acute care hospital (16.4% in 1997 to 2000 vs. 29.8% in 2004 to 2006). One-year mortality after long-term acute care hospital admission was high throughout the study period: 50.7% in 1997 to 2000

and 52.2% in 2004 to 2006.²⁷ Similar data were recently reported in Europe: Over a period of 15 years, there were increases in the number of comorbidities per patient (from 1.8 to 3) and the duration of previous ICU stay (from 25 to 32 days). Given, the greater severity of disease among patients, the overall weaning success rate decreased from 87% to 66%, and the discharge destination changed over time: Fewer patients were discharged to home (decrease from 22% to 10%) and more were discharged to chronic care facilities (increase from 3% to 6%), acute hospitals (increase from 6% to 10%) and rehabilitative units (increase from 70% to 75%). The mortality rate also increased over time (from 9% to 15%).

A total of 1419 patients were enrolled in the United States for the Ventilation Outcomes Study. Patients were old (i.e. >75 years) and averaged 6.9 procedures and treatments during hospitalization with a median length of stay of 40 days (range: 1 to 365 days). The patients had very poor functional status at admission; nearly all were totally bedridden as a result of prolonged critical illness. The prevalence of penetrating or indwelling catheters, each breaching host defenses against infection, was striking. Discharge disposition included 28.8% to home, 49.2% to rehabilitation and extended care facilities, and 19.5% to short-stay acute care hospitals.¹⁰⁶

Bigatello et al¹⁴ noted that many patients are able to be separated from the ventilator at the time of admission to a CVF, which implies that discontinuation of mechanical ventilation was not a top priority in the referring ICU. Their data show that a number of patients may wean rapidly once ventilator management becomes the focus of care in a specialized CVF, confirming a similar observation by Vitacca et al¹⁰⁷ in patients with chronic obstructive pulmonary disease (COPD) and a tracheostomy.

Table 33-5 summarizes the outcome indices for patients admitted to nonacute settings for weaning. There is overall agreement that a rehabilitation-based weaning unit can assist with weaning and maximal functional independence and prepare the family for the discharge of the ventilated patient to home.⁷⁴

Table 33-5: Outcomes for Ventilator-Dependent Patients Admitted to a Chronic Ventilatory Facility Outside an Acute Care Hospital

Auhor (Ref)	Year	Patients (n)	Age (years)	Patients	Patients Weaned (%)	Patients Died in Hospital (%)	Patients Died within 1 Year (%)	Duration of MV (days)	Postdischarge Location (%)
Indihar ⁷¹	1991	171	60	Mixed	34%	60%	?	55	?
Freichels ¹⁰⁸	1993	442	?	Mixed	?	31.7%	?	48	H = 20%; CVFo = 35% Other = 45%
Nava ⁷⁸	1994	42	67 ± 9	COPD	55%	29%	35%	44	?
Scheinorn ⁷²	1994	421	68 ± 0.9	Mixed	74%	28%	72%	?	H = 50% Other = 50%
De Vivo ¹⁰⁹	1995	435	40	SCI	?	?	25%	?	?
Bagley ⁷⁴	1997	278	67	Mixed	38%	31%	?	(11 to 75)	H = 29% CVFo = 71%
Scheinorn ⁷³	1997	1123	69 ± 3	Mixed	56%	29%	62%	29 (1 to 226)	?
Escarrabil ¹¹⁰	1998	10	59 ± 8	ALS	0%	10%	70%	90 (15 to 150)	H = 100%
Votto ⁷⁵	1998	293	(45 to 70)	Mixed	?	?	29% to 60%	?	?
Carson ⁸⁰	1999	133	71	Mixed	70% (discharged) 38% (total)	50%	77%	?	H = 18%; CVFo = 63%; Other = 19%
Modowal ¹¹¹	2002	145	66 ± 16	Mixed	50%	34%	?	94 ± 82	?
Schonhofer ⁶³	2002	403	66	Mixed	68%	24%	51%	41	H = 28% CVFo = 16% Other = 56%
Stoller ¹¹²	2002	162	65 ± 14	Mixed	?	17%	57%	?	H = 28% CVFo = 63% CVFi = 9%
Ceriana ¹¹³	2003	40	67 ± 12	Mixed	67%	15%	?	?	?

Auhor (Ref)	Year	Patients (n)	Age (years)	Patients	Patients Weaned (%)	Patients Died in Hospital (%)	Patients Died within 1 Year (%)	Duration of MV (days)	Postdischarge Location (%)
Lindsay ⁸⁷	2004	102	?		67%	20	?	?	H = 36% NH = 20% Other = 44%
Pilcher ⁶²	2005	153	62 (49 to 72)	Mixed	38% (19% NM; 56% COPD)	28% (15% NM; 50% surgical)	42%	?	?
O'Connor ²²	2009	135	74 (36 to 91)	Mixed	43%	37%	37%	<37	?
Chadwick ¹¹⁴	2009	30	66 ± 8	Motoneuron disease	53.4%	10%	56.7%	34 ± 24	H = 57%
Polverino ¹¹⁵	2010	3106	76 ± 4	Mixed	87 to 66%	9% to 15%	13% at 3 months	<28	H = 16% NH = 3.7% Rehabilitation = 72%

Abbreviations: ALS, amyotrophic lateral sclerosis; COPD, chronic obstructive pulmonary disease; CVFi, chronic ventilator facility inside an acute care hospital; CVFo, chronic ventilator facility outside an acute care hospital; H, home; MV, mechanical ventilation; NH, nursing home; NM, neuromuscular; SCI, spinal cord injuries; (), Range.

Clinical outcome of patients requiring prolonged ventilation depends on the underlying disease. Weaning success is highest in postoperative patients (58%) and patients with acute lung injury (57%), and lowest in patients with COPD or neuromuscular disease (22%).⁷⁴ Chronically ventilated patients with respiratory failure caused by COPD have a worse prognosis than patients with respiratory failure from other causes.⁷⁵ This observation is consistent with the findings of Schonhöfer et al⁶² that long-term survival rate was worse in patients with severe COPD than in other patients. In a homogeneous group of forty-two patients with COPD, Nava et al⁷⁸ observed a successful weaning rate of 55% when a rehabilitation program was continued for a long period outside the ICU.

Physical function is limited and reduced in most patients^{60,116,117} but it is sometimes good or improved after discharge.^{60,80,111,117} Quality of life is defined as good, quite good, reasonable, or normal, although severe impairment is reported in a minority of studies;^{60,80,111,117} often it is improved 1 year after discharge.¹¹⁸ Ambrosino et al¹¹⁷ conducted a prospective, controlled cohort study in a respiratory intermediate ICU in sixty-three patients with COPD requiring mechanical ventilation. Perceived health status and cognitive function were worse in patients recovering from acute-on-chronic respiratory failure (requiring mechanical ventilation) than in stable patients receiving long-term oxygen therapy who never required ICU admission. After discharge, cognition and mobility improved to levels found in stable COPD patients on oxygen therapy.

Problems Unique to Chronic Facilities

Finances

Comparison between studies of different centers and different periods may be inappropriate because weaning success seems strongly related to patient complexity and comorbidities, hospital organization and personnel expertise, availability of early physiotherapy, use of

weaning management techniques, patient autonomy, and family preparation for home discharge with a ventilator. Published data based on small samples, different clinical histories, costs based on patients in a single hospital, different reimbursements, variations in care habits among different countries, and differences in interventions, equipment, and staff involvement reduce the generalizability of cost assessment. In all studies, lower costs principally arise from a lower staffing ratio; other reductions are related to decreased room charges, lower overheads, simpler (usually noninvasive) monitoring, and changes in the pattern of diagnostics and therapeutics.

Costs are likely to change across years, particularly for admissions and related provisions because reimbursement based on DRG does not necessarily reflect real costs of individual treatment. In this respect, CVFs have a peculiar system of reimbursement that deserves special attention.¹¹⁹ As shown in Table 33-6, most observational studies estimate that the daily cost of care for ventilator-dependent patients is lower in a CVF than in an ICU. Indeed, the difference in health care systems between the United States and most other industrialized countries (i.e., near universal health insurance coverage)¹²² makes any comparison between different countries futile.

Table 33-6: Costs Per Day for a Ventilator-Dependent Patient Admitted to an Intensive Care Unit or to a Chronic Ventilator Facility

Authors (Ref)	Year of Publication	Year of Analysis	Type of Unit	Daily Costs
Sheinhorn ⁷²	1994	?	WC (outside acute care H)	\$980
Latriano ⁵⁸	1996	?	Nonmonitored care floor (inside acute care H)	\$453
Bagley ⁷⁴	1997	1995	WC (outside acute care H)	\$630
Nava ¹²⁰	1997	1995	RICU (outside acute care H)	\$865
Gracey ⁵⁶	2000	1998	CVDU (inside acute care H)	\$1084
Engoren ¹¹⁶	2000	?	Cardiac stepdown unit (inside acute care H)	\$439
Seneff ¹⁰³	2000	?	ICU	\$4174
Lindsay ⁸⁷	2004	?	Nursing home	\$303
Halpern ¹	2004	2000	Daily cost per patient admitted to a critical care bed	\$2647
Pilcher ⁶²	2005	1997 to 2000	Cost per day	€1350
O'Connor ²¹	2009	2002 to 2003	Cost per day	\$1054
Carpene ¹²¹	2010	2008 to 2009	Mean cost saving per patient	€39.8452
Kahn ²⁷	2010	1997 to 2006	Mean cost per patient	\$21.766

Note: Reference 1 is the actual daily cost of a critical care bed in year 2000 in the United States.

Abbreviations: CVDU, chronic ventilator-dependent unit; H, hospital; RICU, respiratory intensive-care unit; WC, weaning center.

Proposed modalities of reimbursements in the United States include: (a) single payment for all patients within a DRG with specific weight severity; (b) payment according to variability of units dedicated to prolonged mechanical ventilation; (c) reduced payment for short length of stay; (d) ability to balance with very long stay costs; and (e) bonuses for unweanable patients.

Familiarity with correct reimbursement codes for documentation of time spent caring for patients and appropriate documentation has been advocated for physicians involved in CVFs¹²³ because of the complexity of the encounter with subsequent different payments. To address the problem of underreimbursement, some CVFs are now licensed as DRG-grouping-exempt,⁵⁶ and are required by the Health Care Financial Administration to maintain a mean length of stay more than 25 days and usually fewer than 90 days. In the United States, long-term acute care facilities are reimbursed under regulations of the Tax Equity and Fiscal Responsibility Act (TEFRA)¹²⁴ for care provided to Medicare patients. Charges are reimbursed up to the annual maximum cap for the facility, calculated during a 12-month period designated as the base year. Hospitals incur penalties when charges for Medicare exceed the discharge target amount and receive incentives if charges are reduced in the subsequent years. This policy, unfortunately, has led, as it did for rehabilitation hospitals, to substantial extra costs, including increases in payments to hospitals and doctors and numbers of hospital days for the average patient.¹²⁴ In most European countries, the health care system is funded primarily by government. Therefore, the vast majority of public hospitals and some private hospitals receive most of their funds through a national health service. Most beds are devoted to the treatment of acutely ill patients, independent of the baseline disease (i.e., medical or surgical), and are reimbursed through a DRG-based system. A minor share is devoted to the care of chronically ill patients. The latter beds are located inside rehabilitation wards of acute care hospitals or within independently structured rehabilitation hospitals. For acute care, the DRG-based reimbursement *per case* is applied. For chronic care, reimbursement is on a *per diem* basis, allowing for some increase according to DRG classification. This *per diem* fee applies for only a limited number of days (40 to 60), after which the fee is curtailed drastically.

Mean duration of stay in CVFs differs considerably between patients with an impressively high standard deviation. Nasraway et al³⁷ reported the duration of hospital stay to range from 1 to 2125 days. Because the time spent in CVFs differs so greatly, it is clear that exemption from the DRG-based payment system is granted to avoid massive losses. The *per diem* reimbursement up to relatively small ceiling of days, however, may not achieve reasonable reimbursement.

Staffing

CVFs, especially those outside acute care hospitals, are still characterized by heterogeneous staffing. Most of the centers share common views about the equipment to be used (i.e., ventilators and monitoring systems) and the overall multidisciplinary organization, aimed at improving the autonomy of patients, privacy, and environment. There is, however, considerable discrepancy concerning the training of personnel, especially of nurses. The availability of skilled nurses in extended care facilities in the Boston area, as determined by an informal survey,³⁷ Marked differences exist in the care that each center provides to long-term ventilated patients. The authors³⁷ state that “some facilities may accept ventilator-dependent patients but may not be able to provide for all their needs, especially when serious infections or other setbacks ensue.” Nurses working in a CVF should be specifically trained not only in acute lifesaving procedures, such as resuscitation, but also in specific “chronic” procedures, such as bronchial toilet, prevention of sores at the tracheotomy site, and positioning of the tracheal cannula.

Vitacca et al¹²⁵ studied the allocation of nursing time in a respiratory unit belonging to a rehabilitation center. In the first 2 days after admission, time devoted to care of a ventilator-dependent patient consumed approximately 45% of a nursing shift.

Respiratory therapists should be skilled in weaning protocols that shorten the duration of mechanical ventilation. The hospital team should be trained in clinical tests (i.e., bronchoscopy, fluoroscopy, assessment of respiratory muscle function) that may help the clinicians decide whether or not a patient can have the tracheotomy removed.²¹ Special attention should be paid to the diagnosis of ICU-acquired neuromuscular abnormalities, which increase the time to weaning. Diagnosis may require electrophysiologic studies and needle electromyography of the limb and respiratory muscles.^{126,127}

The global economic crisis in recent years may induce administrators to reduce the number of hospital employees, especially outside critical care settings. In the past, however, a reduction of hospital staff caused a worsening in clinical performance. For example, studies show a lower mortality and length of stay with high-intensity physician staffing.¹²⁸ Dara and Afessa observed a longer stay during the periods with less intensivists in a medical ICU.¹²⁹ Differences in the intensivist-to-ICU-bed ratio, ranging from 1:7.5 to 1:15, were not associated with differences in ICU or hospital mortality, although a ratio of 1:15 was associated with an increased ICU length of stay.¹²⁹ An Italian multicenter study revealed a progressive decrease in the doctor-to-patient ratio secondary to a reduction of medical personnel consequent to the recent restriction in the reimbursement policy of the National Health Care System.¹¹⁵ The negative influence of

reduced medical staff availability on weaning success rate, home discharge, and length of stay has been previously demonstrated,¹¹⁵ providing empirical evidence for a link between organization and outcome in dedicated weaning facilities.

Conclusion

Over the past 15 years, the availability of ICU beds, new technology, and improved levels of care have produced a new population of patients termed *survivors of catastrophic illness*. These patients commonly require prolonged weaning.

The rate of achieving complete ventilator independence in specific and dedicated weaning units is generally high. It has been demonstrated that these units are cost-saving alternatives to an ICU for carefully selected patients. Survivors have an acceptable long-term quality of life. Weaning success, however, does not in itself solve other severe problems, such as the heavy financial and human burdens that the high level of dependency imposes on families, caregivers, and health service organizations once these patients are discharged from a protected environment.

Long-term acute care hospitals play an increasingly important role in patients with chronic critical illness. Yet few data exist to guide decision making about transfer or to inform policy decisions about whether to support or restrict this rapidly growing cost center.²⁷ The different international medical systems need to adopt new organizational innovations and highlight the need for a diverse program of comparative effectiveness research to determine the optimal organization of care for patients recovering from critical illness, including the best way to maximize survival and control costs for this high-risk patient group.²⁷ For these reasons, rigorous studies on structural factors relating to the outcome of patients with mechanical ventilation are mandatory. With increased efforts to reduce health care costs, patients will be shifted away from ICUs toward other clinical settings, such as dedicated weaning facilities, to care for more, and increasingly complex, patients.¹³⁰ It is imperative that we critically evaluate these changes as they occur. More research is needed on the impact of weaning facilities on costs and outcomes. In particular, the following questions should be assessed: (a) whether weaning facilities best operate as units within hospitals or as stand-alone hospitals; (b) the optimum nurse, physician, and ancillary staffing of weaning facilities; and (c) the spillover effects of weaning facilities on critical care.¹³⁰

In conclusion, the main benefits of chronic ventilator facilities are (a) the possibility of relieving congestion of ICU beds, (b) maintaining a high level of nursing assistance, (c) responding to sudden changes in a patient's clinical condition, (d) allowing enough time for a multidisciplinary rehabilitation approach, and (e) acting as a bridge to home-care programs or other forms of continuous chronic assistance (e.g., telemedicine or dedicated long-term units).

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